Oskar Zielinski

LinkedIn | oskarmzielinski@gmail.com | oskarzielinski.dev

MSc Microelectronics and Systems Design student with experience in SystemVerilog RTL, FPGA validation, simulation, fixed-point arithmetic, DSP datapath design, processor-style hardware, CMOS layout, and Python programming. Interested in digital hardware roles spanning GPU datapath design, arithmetic hardware, RTL implementation, verification, and silicon engineering.

Education

2025 – 2026 **MSc Microelectronics and Systems Design**, University of Southampton
Modules: Digital Systems Design, VLSI Systems Design, VLSI Design Project, Embedded Processors, Digital Systems Synthesis

2021 – 2025 **BEng Electrical and Electronic Engineering**, Liverpool John Moores University — *First Class*
Modules: Further Electronic Design, Engineering Project, Engineering Management, Embedded Systems Programming

Work Experience

CE&I Design Engineer Placement - Sellafield — June 2023 - September 2024

- Updated and maintained CE&I drawing packs in AutoCAD (schematics, wiring, panel layouts), implementing approved redline markups into controlled revisions to support design changes and as-built accuracy.
- Interpreted electrical schematics and connection diagrams to support equipment replacement and integration activities; ensured compatibility and correct like-for-like substitution of COTS components.
- Produced an internal document on cybersecurity safety considerations for site wide use, summarising threats/controls and aligning guidance with site engineering practices.
- Designed an electrical panel enclosure for a new building egress/ingress system: defined I/O and power needs, selected components, produced drawings, and documented interfaces for installation/commissioning.
- Coordinated with stakeholders (design, SMEs, other disciplines) to resolve comments and progress drawing/document approvals.

Projects

Weather Station ASIC (SystemVerilog / FPGA group project)

- Implemented SystemVerilog control logic for a 16-character LCD interface, including initialisation, command/data transfer, busy-flag handling, and handshaking for an ST7066U-style parallel display.
- Implemented display-data preparation for rainfall and wind-speed outputs, formatting sensor-derived values into LCD-ready character data.
- Wrote SystemVerilog testbenches and stimulus to verify character writes, control sequencing, address handling, reset behaviour on new data, and correct LCD read/write operation.
- Used targeted testbenches to debug an unexpected extra rain pulse, tracing the issue methodically across calibration and rain-gauge logic before identifying a transient pulse caused by button-synchronisation flip-flop behaviour.
- Validated the LCD subsystem on Intel FPGA hardware, confirming correct display output.
- Assigned and tracked tasks, organised regular team meetings to review progress, resolve blockers, and keep the project aligned with deadlines.

FFT Butterfly Processor (SystemVerilog / FPGA individual project)

- Designed and implemented a SystemVerilog FFT butterfly core for an 8-point FFT application, taking the design from algorithmic specification to RTL and FPGA demonstration.
- Performed manual scheduling and resource binding, converting operations into a clocked controller/datapath architecture with an FSM and registered datapath.
- Implemented signed fixed-point DSP arithmetic, including 8-bit twiddle-factor encoding, 16-bit intermediate multiplication, and truncation back to 8-bit signed outputs.
- Optimised arithmetic resource usage by sharing multiplier resources across sequential states, reducing hardware cost through schedule aware datapath design.
- Verified arithmetic correctness and control sequencing in simulation, then demonstrated handshake driven data capture and output on Intel FPGA hardware.

Application Specific Embedded Processor for FPGA-Based Gaussian Smoothing

- Designed a small 8-bit application-specific processor in SystemVerilog for Gaussian smoothing of signed waveform samples on an FPGA.
- Defined a minimal instruction/control sequence to poll switch inputs, read five ROM samples around the selected index, multiply by fixed kernel coefficients, accumulate the result, and drive the LED output.
- Built the datapath around signed 8-bit arithmetic, 16-bit multiplication, address generation for (W[i-2]) to (W[i+2]), and fixed-point result extraction after coefficient scaling.
- Reduced hardware cost by keeping the architecture specific to the convolution task, avoiding unnecessary general-purpose instructions, registers, memory, and output-control logic.
- Verified the design using SystemVerilog testbenches, full-system simulation in XCelium/SimVision, Quartus RTL/synthesis analysis, and FPGA hardware demonstration.

8 Bit CPU Undergraduate final year project

- Designed and simulated a complete 8-bit Von-Neumann CPU in Logisim Evolution, integrating ALU, registers, FSM control and a control ROM to drive a fixed Fetch–Decode–Execute cycle.
- Verified correct arithmetic, memory access, branching and conditional halting through single-step simulation, DIP-switch stimulus and a full looping program, using predefined test cases to debug and verify correctness.
- Documented expansion roadmap including assembler toolchain, interrupt/stack support and UART I/O.

Magic Transistor Design (TSMC 180 nm) — Cell Library Project

- Designed a full-adder and other minor cells in Magic using two metal layers, adhering to DRC and enforcing I/O and height for library integration.
- Implemented transistor-level schematics from Boolean equations (incl. compound gates), produced stick diagrams, and iterated layout to reduce area and interconnect (metal/poly) lengths.
- Ran parasitic extraction (ext2spice) and automated delay/capacitance sweeps across load and input-slew corners, generating databook tables for each input–output pair.
- Coordinated with team members to ensure cell dimensions, I/O placement, and layout conventions were aligned for integration into the final bit-slice cell library.

EMG-Controlled Rehabilitation System (Medical Electronics group project)

- Contributed to the development and testing of a two-sensor forearm EMG rehabilitation system linking muscle activity acquisition to robotic-hand actuation and a software-based training demonstrator.
- Participated in forearm sensor placement and reproducibility testing, evaluating finger-specific signal signatures and identifying trade-offs in repeatability and signal separation.
- Contributed to EMG preprocessing and decision logic, including filtering, RMS-based feature extraction, and threshold classification, to convert two-channel EMG signals into gesture commands.
- Helped integrate processed EMG outputs with a robotic hand through Python–Arduino serial communication and servo control.
- Contributed to the development of a finger-training game intended to support rehabilitation by encouraging repeated voluntary finger movement in an interactive format.

RC/Line Follower Vehicle

- Built an Arduino-based line follower using IR line sensors with ultrasonic obstacle detection for navigation.
- Implemented embedded control logic to smooth steering, handle sensor dead zones, and stabilise tracking.
- Designed a custom remote control and compared Bluetooth vs RF links; selected RF for more reliable range, latency and sensor ease of use.

Skills

- **RTL / Logical Design:** SystemVerilog, RTL development, FSM/datapath design, fixed-point arithmetic, processor design
- **Verification / Validation:** Testbenches, simulation, waveform debugging, FPGA validation, Xcelium/SimVision
- **Physical Design Exposure:** CMOS layout, standard-cell layout, DRC, parasitic extraction, delay/capacitance characterisation
- **Tools / Programming:** Vivado, Quartus, Magic, Python, Tcl usage
- **Workflow:** Linux, CLI workflow, VS Code, LaTeX, Microsoft Office